

DESI CHHORA

Python AI Assistant

Project Report

1. Introduction

Desi Chhora is a Python-based command-line assistant that brings several interactive utilities into one menu-driven program. The project demonstrates fundamental Python programming concepts through practical features.

2. Problem Statement

A beginner-friendly console assistant is developed to provide basic conversation and common utility functions from a single application. The program uses a simple numbered menu so that a user can select and operate different features.

3. Objectives

The objectives are to create an interactive Python application, practice conditional statements and loops, process user input, perform calculations, implement simple algorithms, and demonstrate date/time handling.

4. Functional Requirements

The application provides seven functional areas: Baat-Cheet, Calculator, Prime Number Check, Maximum/Minimum, Quiz, Date & Time, and Help. An eighth menu option exits the application.

5. Non-Functional Requirements

Usability: the application uses a simple numbered console menu. Reliability: calculator input includes basic ValueError handling and division-by-zero protection. Maintainability: the program separates features into clear menu branches. Resource Efficiency: it is a lightweight command-line application with no external database or network dependency.

6. System Architecture

The application follows a simple menu-driven architecture. User input is received through the console, the selected menu option routes execution to the relevant feature, the feature processes the input, and the result is printed back to the console.

7. Process / Workflow

Start → Display welcome message → Display menu → Read choice → Execute selected feature → Display result → Return to menu → Exit when option 8 is selected.

8. Implementation Details

The project uses Python's input/output facilities, conditional statements, while loops, for loops, list processing, arithmetic operations, string matching, exception handling, and the datetime module. The prime-number check tests divisors up to the square root of the number. The date/time feature uses datetime.now() and strftime().

9. Features

Baat-Cheet responds to selected greetings and common phrases. Calculator performs +, -, * and / operations. Prime Number Check identifies whether a number is prime. Maximum/Minimum finds the largest and smallest values from entered numbers. Quiz asks three questions and displays the score. Date & Time displays the current date and time. Help lists available capabilities.

10. Testing Approach

Testing should cover every menu option, valid and invalid inputs, calculator operations, division by zero, prime and non-prime values, maximum/minimum inputs, correct and incorrect quiz answers, date/time output, invalid menu selections, and program exit.

11. Challenges Faced

The project requires handling continuous menu interaction, matching user text for basic conversation, validating numeric input, preventing division by zero, and implementing prime-number and maximum/minimum logic.

12. Learnings & Key Takeaways

The project provides practice with Python syntax, loops, conditions, user input, exception handling, lists, string methods, arithmetic operations, and the datetime module. It also demonstrates how multiple small features can be integrated into one interactive program.

13. Future Enhancements

Future versions can include more conversation patterns, additional calculator operations, a larger quiz question bank, score history, persistent data storage, modular source files, and a graphical user interface.

14. References

Python standard library concepts used in the project, particularly the datetime module. The project source code supplied for this report is the primary implementation reference.